

IMPACT OF DYNAMIC LINE RATING ON THE GERMAN ELECTRICAL ENERGY SYSTEM

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ABSTRACT

The deployment of new renewable energy sources far from load centers leads to an increasing need for grid expansion. A notable example can be found in Germany, where most wind energy is generated in the north and has to be transmitted to the industrial south [1]. In contrast to the planned increase in wind power capacity of 130 GW until 2030 by the current German government, only the minority of grid expansion projects are under construction [2]. Moreover, most of these projects have been delayed due to administrative problems and protest activities against grid expansion[3]. This trend will likely narrow the bottleneck of the transmission system, which can lead to more grid instabilities and curtailment.

The transmission capacity of an overhead transmission line directly depends on its ampacity. Traditionally, transmission line ampacity is calculated at unfavorable, static weather conditions [4], referred to as Static Line Rating (SLR). By design, SLR underestimates the ampacity of a transmission line. When implementing in the real world, SLR leads to an overestimated transmission capacity expansion. In contrast, Dynamic Line Rating (DLR) considers time varying weather conditions to calculate the transmission line ampacity. Therefore, the implementation of DLR has several benefits such as cost-efficiency, congestion reduction and better wind power integration [5]. In literature, case studies consider DLR in simulations [4] or in small scale Energy System Optimization Models (ESOMs) [6] [7] [8]. However, no case study evaluates the impact of DLR in a higher scale ESOM e.g. national scale.

In this paper, we analyze the impact of DLR on the German power system. For this, we build an ESOM based on the PyPSA-Eur framework [9]. The model has 40 nodes, is hourly resolved and considers a time horizon of one year. To calculate DLR in our model, we use the IEEE standard [10] with highly resolved historical weather data using *atlite* [11]. With our model, we compare the German power system applying SLR and DLR. For the comparison, we analyze the wind curtailment, the grid congestion as well as the generator deployment of both ESOMs. This allows us to derive the benefits of DLR in the German power system.

Preliminary results show a cost reduction of 5-10 % in the ESOM applying DLR compared to SLR. Furthermore, DLR leads to approx. 15 % higher wind power deployment and approx. 75 % higher average power flow compared to SLR.

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